

Methods of solving combinatorial problems

Four working methods — the tree, the table, the rules, and counting the opposite case.

LESSON 161–163

3 × 40 MIN

MATEMATIKA 7, §67

S8 13

LESSON CLOCK

15'

30'

30'

30'

15'

Choosing a method	15 min
Dealing with a restriction first	30 min
Counting the opposite case	30 min
Worked problems	30 min
Checking an answer	15 min

LEARNING OBJECTIVES

- Choose a method that fits the size of the problem.
- Handle a restriction by dealing with it first.
- Count a difficult case as the total minus the opposite one.
- Check an answer by a second method or by a small case.

TEXTBOOK

National	pp. 484–494
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.3

01

Explanation

Choosing a method

METHOD	USE IT WHEN	DANGER
a systematic list	under about 20 outcomes	missing one
a tree diagram	two or three stages	the page runs out
the two rules	most problems	choosing the wrong rule
the opposite case	“at least one” problems	forgetting to subtract from the total

LIST SMALL, CALCULATE LARGE

A list is worth writing when it can be finished in a minute — and it is then the best check on a calculation. Beyond that, the rules are the only practical route.

Dealing with a restriction first

Whatever is restricted is chosen at the first stage; the free choices follow.

PROBLEM	FIRST	THEN	ANSWER
3-figure numbers from 1,2,3,4,5, no repeats	any digit: 5	$4 \cdot 3$	60
the same, but even	last digit 2 or 4: 2	$4 \cdot 3$	24
the same, but starting with 5	first digit fixed: 1	$4 \cdot 3$	12
seating 4 pupils with one fixed at the end	that pupil: 1	$3!$	6

CHOOSE THE RESTRICTED PLACE FIRST, NOT LAST

If the units digit is chosen last, the number of choices left depends on what went before and the product is no longer a single multiplication. Taking the restriction first keeps every stage independent.

Counting the opposite case

“At least one” is nearly always easier as “all of them minus none of them”.

at least one = total – none

PROBLEM	TOTAL	NONE	ANSWER
3-figure codes with at least one 7	1000	$9^3 = 729$	271
three coins with at least one head	8	1	7
two dice with at least one six	36	25	11

“AT LEAST ONE” IS NOT “EXACTLY ONE”

Two dice give 11 outcomes with at least one six, but only 10 with exactly one — the double six belongs to the first count and not to the second.

Worked problems

PROBLEM	WORKING	ANSWER
3-figure numbers from 1,2,3,4,5 without repeats	$5 \cdot 4 \cdot 3$	60
how many of them are even	$2 \cdot 4 \cdot 3$	24
how many are odd	$60 - 24$	36
how many are greater than 300	$3 \cdot 4 \cdot 3$	36
a committee of 2 from 4 boys and 3 girls with one of each	$4 \cdot 3$	12

EVEN AND ODD SHOULD ADD TO THE TOTAL

$24 + 36 = 60$ ✓. Splitting a count into two cases gives a free check: the parts must rebuild the whole.

Checking an answer

CHECK	HOW
a small case	redo the problem with 2 or 3 objects and list them
the complement	the case and its opposite must add to the total
a second method	tree against rules, or list against product
size	an answer larger than the total is impossible

THE SMALL CASE IS THE BEST CHECK THERE IS

If a formula gives 45 handshakes for 10 people, test it on 3 people: it should give 3, and three people shaking hands really do make three handshakes.

02 Worked examples

EXAMPLE 1 How many 3-figure numbers with different digits can be made from 1, 2, 3, 4, 5, and how many of them are even?

① $5 \cdot 4 \cdot 3 = 60$ altogether.

② Even: the units digit must be 2 or 4 — 2 ways.
The restriction first.

③ The other two places: $4 \cdot 3 = 12$.

④ $2 \cdot 12 = 24$ even numbers.
Check: $24 + 36 = 60$ ✓

ANSWER

60 in all, 24 even

EXAMPLE 2 How many 3-figure codes from 10 digits contain at least one 7?

① Total: $10^3 = 1000$.

② With no 7: $9^3 = 729$.
Nine digits remain at each place.

③ $1000 - 729$

④ $= 271$

ANSWER

271

EXAMPLE 3 A committee of two is chosen from 4 boys and 3 girls, one of each. How many committees?

① One boy: 4 ways.

② One girl: 3 ways.

③ “And” — multiply.

④ $4 \cdot 3 = 12$

ANSWER

12

03 Interactive model

Ask for the number of three-coin outcomes with at least one head; most of the class starts listing, and the total-minus-none route wins by a distance.

Restrictions and opposites

3-figure numbers from 1,2,3,4,5 without repeats:

15

60

125

120

How many are even?

12

24

30

36

How many are odd?

24

30

36

40

The restricted place should be chosen:

first

last

in the middle

at random

“At least one” is best counted as:

exactly one

total minus none

total plus none

none

3-figure codes with at least one 7:

271

300

729

100

Three coins with at least one head:

3

6

7

8

Two dice with at least one six:

6

10

11

12

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Method	Usul	Метод
Restriction	Cheklov	Ограничение
Complementary counting	Teskarisini sanash	Подсчёт дополнения
At least one	Kamida bitta	Хотя бы один
Systematic list	Tartibli ro‘yxat	Систематический перечень
To check	Tekshirmoq	Проверить
Small case	Kichik holat	Малый случай
Total	Umumiy soni	Общее число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A restriction should be handled:

first

last

never

twice

“At least one” equals:

exactly one

total minus none

none

all

3-figure codes with at least one 7 number:

271

729

100

300

Two dice with at least one six:

10

11

12

6

A good check on a count is:

doing it again the same way

a small case

guessing

rounding

Even and odd counts must:

be equal

add to the total

differ by one

multiply to the total

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3-figure numbers from 1,2,3,4,5, no repeats

ANSWER 60

2. How many are even

ANSWER 24

3. How many are odd

ANSWER 36

4. How many start with 5

ANSWER 12

5. Three coins: outcomes in all

ANSWER 8

6. Three coins with at least one head

ANSWER 7

7. Two dice: outcomes in all

ANSWER 36

Medium · the standard question

7 PROBLEMS

1. 3-figure codes with at least one 7

ANSWER 271

2. Two dice with at least one six

ANSWER 11

3. Two dice with exactly one six

ANSWER 10

4. A committee of one boy and one girl from 4 and 3

ANSWER 12

5. 3-figure numbers from 1,2,3,4,5 greater than 300

ANSWER 36

6. Seating 4 pupils with one fixed at the end

ANSWER 6

7. 4-figure PINs with no 0

ANSWER 6561

Hard · stretch and reason

7 PROBLEMS

1. 4-figure codes with at least one 7

ANSWER 3439

2. Three dice with at least one six

ANSWER 91

3. 3-digit numbers with no repeated digit and even

ANSWER 328

4. Committees of 2 from 7 pupils

ANSWER 21

5. Committees of 2 from 7 with at least one of 3 girls

ANSWER 15

6. Why check with a small case?

ANSWER It can be listed in full and compared

7. Arrangements of 5 books with two particular ones together

ANSWER 48

07 Homework

Homework — 5 tasks

For every “at least one”, count the opposite case and subtract.

- How many 3-figure numbers with different digits can be made from 2, 3, 4, 5, 6?
- How many of them are even?
- How many 4-figure codes from 10 digits contain at least one 9?
- How many outcomes of four coins contain at least one tail?
- A committee of one boy and one girl is chosen from 5 boys and 6 girls. How many committees?